

SHISHMAREF, AK, USA

WMO: 701195

Lat: 66.250N

Lon: 166.089W

Elev: 7

StdP: 14.69

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26625

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-29.2	-25.4	-29.2	1.1	-16.5	-28.3	1.1	-17.3	35.6	0.8	32.2	6.5	8.6	250	0.974

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.9	60.9	55.6	57.3	53.0	55.3	51.6	56.2	60.1	54.0	57.2	52.2	55.3	10.6	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.2	62.4	59.0	52.1	57.8	56.9	49.8	53.1	54.8	23.8	60.0	22.6	57.9	21.4	55.4	62.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
32.0	27.7	24.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			DB												
			WB	-25.7	58.4	2.1	2.3	-27.1	60.1	-28.4	61.4	-29.5	62.8	-31.0	64.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.1	1.0	0.2	-0.7	11.7	29.5	39.8	49.6	48.8	42.9	30.3	17.5	5.3
	DBStd	21.09	14.38	14.25	12.15	11.57	7.39	6.26	4.87	4.36	4.94	6.80	10.24	12.64
	HDD50	9911	1518	1393	1573	1149	637	313	65	74	219	610	975	1385
	HDD65	15282	1983	1813	2038	1599	1102	755	477	501	664	1075	1425	1850
	CDD50	103	0	0	0	0	0	7	53	38	5	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	11.6	12.0	12.4	11.0	10.7	9.0	8.9	10.6	12.0	12.8	12.8	13.5	13.2
Precipitation	PrecAvg	22.8	1.6	1.8	1.2	1.2	1.1	1.4	2.6	3.3	2.3	2.5	2.0	1.6
	PrecMax	30.9	3.6	5.6	3.4	3.1	2.9	2.8	4.8	6.9	4.8	4.4	5.6	2.9
	PrecMin	16.3	0.1	0.4	0.2	0.3	0.2	0.4	0.9	0.9	0.5	0.6	0.3	0.0
	PrecStd	4.0	0.8	1.2	0.9	0.7	0.7	0.5	1.0	1.5	1.0	1.1	1.3	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.3	34.2	33.9	36.4	45.8	60.9	64.4	63.5	56.6	45.4	36.4	35.9
		MCWB	32.5	31.9	31.9	34.0	41.1	55.4	57.3	57.4	51.2	41.4	34.5	32.2
	2%	DB	29.9	30.4	28.4	35.6	40.9	55.3	61.4	59.4	54.1	42.9	34.4	31.9
		MCWB	28.9	29.6	27.1	33.5	37.8	51.1	55.8	54.7	50.5	39.7	32.9	30.8
	5%	DB	26.9	27.0	25.0	33.6	38.7	52.3	58.7	56.9	52.1	40.7	33.6	27.7
		MCWB	25.3	25.3	23.2	31.7	36.1	48.6	54.2	53.1	49.1	38.2	32.0	26.3
	10%	DB	21.6	21.4	18.4	29.7	36.9	48.3	56.6	54.9	49.7	38.8	30.3	23.8
		MCWB	20.4	20.2	17.1	29.0	34.7	45.5	52.4	51.6	47.1	36.5	29.4	22.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.6	32.2	32.0	34.0	41.5	55.9	58.7	58.5	52.3	42.3	34.8	33.0
		MCDB	34.0	33.6	33.4	36.0	45.2	60.0	62.9	62.6	54.7	44.5	35.5	34.6
	2%	WB	28.7	29.8	27.7	32.7	38.2	51.5	56.8	55.9	50.8	40.3	33.0	30.3
		MCDB	29.3	30.6	28.5	34.5	40.7	55.1	60.8	59.5	53.5	42.7	34.2	31.7
	5%	WB	25.4	25.5	23.9	31.5	36.5	48.5	54.8	53.7	49.2	38.5	31.7	26.5
		MCDB	27.2	27.4	25.8	33.2	38.4	51.9	58.4	56.3	51.8	40.5	33.0	27.7
	10%	WB	21.3	21.3	17.3	28.4	35.0	45.9	52.7	52.0	46.8	36.9	29.5	22.9
		MCDB	22.4	22.4	18.5	28.9	36.9	48.5	56.1	55.0	48.8	38.7	30.2	24.2
Mean Daily Temperature Range	5% DB	MDBR	11.1	12.7	14.0	13.5	8.7	8.4	7.9	6.9	6.1	5.9	8.9	10.8
		MCDBR	12.2	16.4	16.2	10.1	10.3	12.7	11.6	9.9	8.5	6.7	6.5	13.5
		MCWBR	12.0	16.0	15.5	9.1	8.4	9.2	7.8	7.0	6.0	5.6	6.4	13.1
	5% WB	MCDBR	12.5	16.5	16.2	11.2	9.9	12.3	10.5	9.1	7.9	6.6	6.7	12.6
		MCWBR	12.4	16.2	15.6	10.4	8.2	9.0	7.6	6.8	5.8	5.7	6.6	12.3
Clear-Sky Solar Irradiance	taub	0.168	0.245	0.268	0.336	0.369	0.351	0.365	0.384	0.315	0.276	0.364	0.178	
	taud	1.904	2.100	2.182	2.006	2.069	2.400	2.441	2.330	2.513	2.297	2.805	1.905	
	Ebn at Noon	86	195	253	256	260	272	264	242	241	184	83	9	
	Edh at Noon	5	17	26	41	43	32	30	30	20	14	6	1	
All-Sky Solar Radiation	RadAvg	28	217	777	1440	1747	1842	1448	1066	656	232	44	3	
	RadStd	3	26	59	87	195	153	127	113	64	22	6	0	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	+2.02	N/A	+1.98	+1.58	+1.15	N/A	-616	-742	+81	N/A	

Nomenclature: See separate page